Normal Schemes and the Arithmetic Fundamental Group

1 Introduction

Throughout these notes, S denotes a connected normal Noetherian scheme with function field $K = k(S)$.

For every connected finite étale covering $X \longrightarrow S$, the function field $k(X)$ is a finite separable extension of K. It is natural to ask which finite extensions of K arise in this way.

Conversely, given a finite separable extension L/K, let $\nu_L : S_L \longrightarrow S$ denote the normalization of S in L. The main result of this chapter identifies the finite étale coverings of S with those finite separable extensions of K for which the normalization is unramified.

This correspondence leads naturally to a description of the étale fundamental group of a normal scheme as a quotient of the absolute Galois group of its function field. In particular, one obtains the exact sequence relating the geometric and arithmetic fundamental groups.

2 Normalization and unramified extensions

Let L/K be a finite separable extension.

Definition 2.1. The normalization of S in L is the integral closure $\nu_L : S_L \longrightarrow S$ of S inside L.

If $U = \operatorname{Spec} A$ is a non-empty affine open subset of S, then $\nu_L^{-1}(U) = \operatorname{Spec} A_L$, where A_L denotes the integral closure of A in L.

Definition 2.2. A finite separable extension L/K is said to be *unramified over* S if the normalization morphism $\nu_L : S_L \longrightarrow S$ is unramified.

The following finiteness result will be used throughout.

Lemma 2.3 (Lemma 6.12)**.** *For every finite separable extension L/K, the normalization morphism $\nu_L : S_L \longrightarrow S$ is finite.*

Proof. Omitted. □

The remaining issue is therefore to determine when the normalization morphism is flat. This is the only non-trivial ingredient in proving that an unramified extension of K gives rise to a finite étale covering of S.

3 Standard étale neighbourhoods

The local structure of unramified morphisms is described by standard étale morphisms.

Definition 3.1. Let A be a ring. A *standard étale* A-algebra is an algebra of the form $C = (A[T]/(P))_{P'(T)}$, where $P \in A[T]$ is a monic polynomial.

Since $P'(T)$ is invertible in C, the morphism $\operatorname{Spec} C \longrightarrow \operatorname{Spec} A$ is étale.

The following local description will be the key tool in the proof of the main theorem.

Lemma 3.2 (Lemma 6.13). *Let $\phi : X \longrightarrow S$ be an unramified morphism and let $x \in X$.*
Then there exist

- *an affine neighbourhood $U = \operatorname{Spec} A$ of $\phi(x)$,*
- *a standard étale A-algebra C,*

such that, after replacing X by an open neighbourhood of x, there is a closed immersion $X \hookrightarrow \operatorname{Spec} C$ over U.

Proof. Omitted. □

After localizing at the image of x, we obtain a surjective homomorphism of local rings $C \twoheadrightarrow \mathcal{O}_{X,x}$. This observation will be used repeatedly in what follows.

4 Finite étale coverings and unramified extensions

The goal of this section is to prove the converse implication: every finite separable extension of K that is unramified over S arises from a connected finite étale covering.

We begin with an elementary algebraic lemma.

Lemma 4.1 (Lemma 6.14). *Let $A \subseteq B$ be an inclusion of domains with fraction fields $K \subseteq L$, where L/K is finite and separable.*
Assume that B is integral over A. Then there exist finitely many elements $x_1, \ldots, x_n \in B$ whose images form a K-basis of L.

Proof. Choose any K-basis $\alpha_1, \ldots, \alpha_n$ of L.
Since each α_i belongs to the fraction field of B, there exists a non-zero element $a \in A$ such that $a\alpha_i \in B$ for every i.
The elements $a\alpha_1, \ldots, a\alpha_n$ still form a K-basis of L, proving the claim. □

The next lemma provides the flatness needed in the proof of the main theorem.

Lemma 4.2 (cf. Lemma 6.15). *Let L/K be a finite separable extension which is unramified over S.*
Then the normalization morphism $\nu_L : S_L \longrightarrow S$ is flat.

Proof. Since flatness is local, let $x \in S_L$.
Write $A = \mathcal{O}_{S,\nu_L(x)}, \qquad B = \mathcal{O}_{S_L,x}$.
By Lemma 3.2, there exists a standard étale local A-algebra C together with a surjection $C \twoheadrightarrow B$.
Tensoring with the fraction field K yields $C \otimes_A K \twoheadrightarrow B \otimes_A K$.
Since B is contained in the function field L, one has $B \otimes_A K \simeq L$.
At this point one shows that the above morphism is an isomorphism, from which it follows that B is finite projective over A. Consequently B is flat over A, proving the lemma. □

Theorem 4.3 (Theorem 6.16). *The assignment $X \longmapsto k(X)$ induces an equivalence between the category of connected finite étale coverings of S and the category of finite separable extensions of K that are unramified over S.*

Proof. Let $X \longrightarrow S$ be a connected finite étale covering.
Then $k(X)/K$ is finite and separable. Since étale morphisms are unramified, the extension is unramified over S.
Conversely, let L/K be finite separable and unramified over S.
By Lemma 2.3, the normalization morphism $\nu_L : S_L \longrightarrow S$ is finite.

By Lemma 4.2, it is flat.

Since it is unramified by definition, it follows that ν_L is finite étale.

Finally, the two constructions are inverse to each other because a connected normal scheme is determined by its function field together with its integral closure over the base. □

5 The fundamental group of a normal scheme

Fix a geometric generic point $\bar{\eta} \longrightarrow \operatorname{Spec} K$, where Ω denotes the corresponding separable closure of K.

By Theorem 4.3, every connected finite étale covering of S corresponds to a finite subextension of Ω/K that is unramified over S. This immediately yields the following description of the étale fundamental group.

Let $M_{K,S}$ be the compositum inside Ω of all finite extensions of K that are unramified over S.

Proposition 5.1 (Proposition 7.1)**.** *There is a canonical isomorphism* $\pi_1(S, \bar{\eta}) \simeq \operatorname{Gal}(M_{K,S}/K)$. *Equivalently,* $\pi_1(S, \bar{\eta}) \simeq \operatorname{Gal}(\Omega/K)/\operatorname{Gal}(\Omega/M_{K,S})$.

Proof. By Theorem 4.3, finite quotients of $\pi_1(S, \bar{\eta})$ correspond precisely to finite Galois extensions of K contained in $M_{K,S}$.

Passing to inverse limits gives $\pi_1(S, \bar{\eta}) \simeq \varprojlim_{L \subseteq M_{K,S}} \operatorname{Gal}(L/K) = \operatorname{Gal}(M_{K,S}/K)$.

Since $M_{K,S} \subseteq \Omega$, the restriction homomorphism $\operatorname{Gal}(\Omega/K) \longrightarrow \operatorname{Gal}(M_{K,S}/K)$ is surjective with kernel $\operatorname{Gal}(\Omega/M_{K,S})$, which proves the second description. □

Consequently, the canonical morphism $\operatorname{Spec} K \longrightarrow S$ induces a surjective homomorphism $\pi_1(\operatorname{Spec} K, \bar{\eta}) \twoheadrightarrow \pi_1(S, \bar{\eta})$.

Since $\pi_1(\operatorname{Spec} K, \bar{\eta}) \simeq \operatorname{Gal}(\Omega/K)$, Proposition 5.1 identifies its kernel.

Corollary 5.2. *There is a canonical short exact sequence* $1 \longrightarrow \operatorname{Gal}(\Omega/M_{K,S}) \longrightarrow \pi_1(\operatorname{Spec} K, \bar{\eta}) \longrightarrow \pi_1(S, \bar{\eta}) \longrightarrow 1$.

Proof. This is immediate from Proposition 5.1. □

6 Geometric and arithmetic fundamental groups

Assume that S is geometrically connected over a field k.

Choose a separable closure k_s/k and write $S_{\bar{k}} = S \times_k k_s$.

Let $\bar{\eta} \longrightarrow S_{\bar{k}}$ be a geometric generic point.

The structural morphism $S \longrightarrow \operatorname{Spec} k$ induces a continuous homomorphism $\pi_1(S, \bar{\eta}) \longrightarrow \operatorname{Gal}(k_s/k)$.

The following exact sequence relates the arithmetic and geometric fundamental groups.

Theorem 6.1. *There is a natural exact sequence of profinite groups* $1 \longrightarrow \pi_1(S_{\bar{k}}, \bar{\eta}) \longrightarrow \pi_1(S, \bar{\eta}) \longrightarrow \operatorname{Gal}(k_s/k) \longrightarrow 1$.

Proof. The surjectivity of the right-hand map follows from the fact that every finite separable extension of k gives rise to a finite étale covering of $\operatorname{Spec} k$.

The kernel consists precisely of those finite étale coverings that become trivial after base change to k_s. Equivalently, these are the finite étale coverings of the geometric fibre $S_{\bar{k}}$, which are classified by $\pi_1(S_{\bar{k}}, \bar{\eta})$.

Passing to inverse limits yields the exact sequence. □

The profinite group $\pi_1(S_{\bar{k}}, \bar{\eta})$ is called the *geometric fundamental group* of S, while $\pi_1(S, \bar{\eta})$ is called the *arithmetic fundamental group*.

The preceding theorem shows that the arithmetic fundamental group is obtained from the geometric fundamental group by adjoining the arithmetic information encoded by the absolute Galois group of the base field.

Example 6.2. If $S = \operatorname{Spec} k$, then $\pi_1(S) = \operatorname{Gal}(k_s/k)$.

On the other hand, $S_{\bar{k}} = \operatorname{Spec} k_s$, whose finite étale coverings are trivial. Hence $\pi_1(S_{\bar{k}}) = 1$, and the exact sequence of Theorem 6.1 reduces to $1 \longrightarrow 1 \longrightarrow \operatorname{Gal}(k_s/k) \longrightarrow \operatorname{Gal}(k_s/k) \longrightarrow 1$.

References

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